Official College Board Digital SAT Question Bank — Released Questions

Predicted for the August SAT — CLC Educational Institute

Problem-Solving and Data Analysis

Math | 19 questions with full Rule / Tip / Trap / POE / Desmos explanations

The CLC Method

Every question in this file is broken down the same way, so you build one consistent habit instead of guessing question to question:

The Rule — the underlying math principle or formula being tested. Learn the rule, not just this one answer, and you'll be able to answer every future question of this type.

The Tip — a concrete move you can make before diving into algebra, so you solve efficiently instead of the long way.

Step-by-Step Solution — every single algebra step written out in full, with nothing skipped — so you can follow exactly how the answer was reached, not just see the final number.

The Trap — the single wrong answer most students actually pick, and the specific arithmetic or logical slip that produces it.

POE (multiple-choice questions only) — a full walkthrough of the other wrong choices, showing exactly what error produces each one. Some Math questions are Student-Produced Response ('grid-in') questions with no answer choices at all — for those, there's no POE section, just the full solution.

Solving with Desmos — exact, specific instructions for using the built-in Desmos graphing calculator (available on every Digital SAT Math question) to solve or verify the question — including the exact expressions to type in, not just a general suggestion to 'graph it.'

Ratios, rates, proportional relationships, and units

3 questions · Easy 1 / Medium 0 / Hard 2

Ratios, rates, proportional relationships, and units — Q1 [Easy] (ID: de799680)

Question ID: de799680

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Easy

Question

A cluster fly is traveling at a velocity of 76 centimeters per second. What is this velocity, in **millimeters** per second?

$(1 \text{ centimeter} = 10 \text{ millimeters})$

Answer

A. 7,600

B. 836

C. 760

D. 86

Correct Answer: C

The Rule: Unit conversion problems always use the given conversion factor as a multiplier — set it up so the unit you want to CANCEL is in the denominator and the unit you want to KEEP ends up in the numerator, then multiply straight across.

The Tip: Write the conversion factor as a fraction in two different orientations (10 mm/1 cm, or 1 cm/10 mm) and pick whichever one cancels out centimeters and leaves millimeters — this prevents accidentally dividing when you should multiply, or vice versa.

Why C is right: Since 1 centimeter = 10 millimeters, converting FROM centimeters TO millimeters means MULTIPLYING by 10 (going from a bigger unit to a smaller one always gives a bigger number). So 76 cm/s × 10 mm/cm = 760 mm/s.

Step-by-Step Solution:

Step 1: Identify the given rate: 76 centimeters per second.

Step 2: Identify the conversion factor: 1 centimeter = 10 millimeters.

Step 3: Since millimeters are a SMALLER unit than centimeters, converting to millimeters should give a LARGER number — this means multiplying, not dividing.

Step 4: Multiply: 76 cm/s × 10 mm/cm = 760 mm/s (the 'cm' units cancel out, leaving 'mm').

Step 5: This matches choice C.

The Trap: Choice D (86) is the most commonly picked wrong answer — it comes from mistakenly ADDING 10 to 76 (76 + 10 = 86) instead of correctly MULTIPLYING (76 × 10 = 760), confusing an additive shift with a multiplicative unit conversion.

POE — Eliminate the other three:

✗ A — 7,600 is off by a factor of 10 from the correct answer — likely from multiplying by 100 instead of 10, perhaps double-applying the conversion factor.

✗ B — 836 doesn't correspond to a natural calculation from this conversion — possibly from adding an unrelated number (760 + 76 = 836) rather than performing the correct single multiplication.

✗ D — 86 comes from mistakenly ADDING 10 to 76 (76 + 10 = 86) instead of correctly MULTIPLYING (76 × 10 = 760) — confusing an additive shift with a multiplicative unit conversion.

Solving with Desmos:

Step 1: Open Desmos and type the conversion directly as a multiplication: 76*10

Step 2: Press Enter — Desmos will instantly evaluate this to 760, confirming the converted velocity in millimeters per second, matching choice C.

Step 3: As a sanity check on direction, type 76/10 as well — this gives 7.6, which would be the (incorrect) result of dividing instead of multiplying; comparing both results helps confirm multiplying by 10 is the correct operation when converting from a larger unit (cm) to a smaller one (mm).

Ratios, rates, proportional relationships, and units — Q2 [Hard] (ID: 4c5ea142)

Question ID: 4c5ea142

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Hard

Question

On average, a certain plant grows 87 millimeters every m months. At this rate, which expression represents the number of millimeters, on average, the plant grows every k years?

Answer

A. $\frac{29m}{4k}$

B. $\frac{29k}{4m}$

C. $\frac{1{,}044m}{k}$

D. $\frac{1{,}044k}{m}$

Correct Answer: D

The Rule: For rate-conversion word problems with two different unit changes happening at once (here, converting a monthly rate to a yearly rate, AND scaling by an unknown number of years), set up the rate as a fraction and multiply by the necessary conversion factors and variables step by step, keeping careful track of what's in the numerator and denominator.

The Tip: Break multi-step unit conversions into separate, small steps rather than trying to do everything at once — first convert 'per month' to 'per year' using the fixed number 12, and only afterward multiply by the variable representing the number of years.

Why D is right: The plant grows 87 mm every m months, so its rate is 87/m mm per month. Converting to a yearly rate: since there are 12 months in a year, multiply by 12: (87/m) × 12 = 1044/m mm per year. For k years, multiply by k: (1044/m) × k = 1044k/m.

Step-by-Step Solution:

Step 1: Write the given growth rate as a fraction: 87 millimeters per m months, or 87/m mm/month.

Step 2: Convert this monthly rate into a YEARLY rate by multiplying by 12 (since there are 12 months in one year): (87/m) × 12 = 1044/m mm per year.

Step 3: To find the total growth over k years (not just one year), multiply this yearly rate by k: (1044/m) × k = 1044k/m.

Step 4: This matches choice D.

The Trap: Choice C is the most commonly picked wrong answer — it correctly computes 1044 (87 × 12) but multiplies by m instead of k, and divides by k instead of m, essentially swapping the roles of the two variables in the final expression.

POE — Eliminate the other three:

✗ A — 29m/4k doesn't come from a clean derivation of this problem — the numbers 29 and 4 don't naturally arise from 87 and 12 in this context (87/3=29, 12/3=4, suggesting a reduction error somewhere, but the variable placement is also backwards).

✗ B — 29k/4m has the same reduction issue as choice A, with variables swapped in a way that doesn't match the correct setup.

✗ C — 1,044m/k correctly computes 1,044 but swaps which variable (m or k) belongs in the numerator versus denominator — m should be in the denominator (since a larger m means slower growth per month) and k in the numerator (since more years means more total growth).

Solving with Desmos:

Step 1: Since this problem has two unknown variables (m and k), pick simple test values to verify your formula — let's say m = 3 (grows 87mm every 3 months) and k = 2 (over 2 years). Step 2: Open Desmos and calculate the rate per month directly: 87/3 — this gives 29 mm per month.

Step 3: Convert to yearly rate: type 29*12 — this gives 348 mm per year.

Step 4: Multiply by the number of years: type 348*2 — this gives 696 mm total growth over 2 years.

Step 5: Now test your formula 1044k/m with the same values: type 1044*2/3 — this should also give 696, confirming your formula matches the step-by-step calculation and verifying choice D.

Ratios, rates, proportional relationships, and units — Q3 [Hard] (ID: 1b96c116)

Question ID: 1b96c116

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Hard

Question

A rectangle has an area of $3{,}168$ square inches. What is the area, in **square feet**, of this rectangle? $(1 \text{ foot} = 12 \text{ inches})$

Answer

A. 22

B. 56.3

C. 264

D. 675.4

Correct Answer: A

The Rule: When converting AREA (not just length) between units, you must apply the conversion factor TWICE — once for each dimension (length and width) — meaning you divide (or multiply) by the SQUARE of the linear conversion factor, not just the factor itself.

The Tip: Never apply a linear conversion factor (like 12 inches = 1 foot) directly to an area measurement — always square the conversion factor first ($12^2 = 144$) before using it to convert square inches to square feet, since area involves two dimensions.

Why A is right: Since 1 foot = 12 inches, 1 square foot = 12 × 12 = 144 square inches. To convert 3,168 square inches to square feet, divide by 144: 3,168 ÷ 144 = 22.

Step-by-Step Solution:

Step 1: Recall the linear conversion: 1 foot = 12 inches.

Step 2: Since area involves TWO dimensions (length × width), square the conversion factor to get the area conversion: 1 square foot = 12^2 = 144 square inches.

Step 3: To convert the given area from square inches to square feet, divide by 144 (since square inches is the smaller unit, converting to the larger unit square feet requires dividing): 3,168 ÷ 144 = 22.

Step 4: This matches choice A.

The Trap: Choice C (264) is the most commonly picked wrong answer — it results from dividing by 12 instead of 144, applying only the LINEAR conversion factor instead of correctly squaring it for an AREA conversion.

POE — Eliminate the other three:

✗ **B** — 56.3 doesn't correspond to a clean derivation from this problem — likely from an unrelated arithmetic error.

✗ **C** — 264 comes from dividing 3,168 by 12 instead of 144 — this uses only the linear (one-dimensional) conversion factor rather than correctly squaring it for the two-dimensional area conversion.

✗ **D** — 675.4 doesn't correspond to a clean derivation from this problem either — possibly from multiplying instead of dividing, or a different unrelated calculation error.

Solving with Desmos:

Step 1: Open Desmos and calculate the area conversion factor first: type 12^2 — this confirms 144 square inches equals 1 square foot.

Step 2: On the next line, type 3168/144 — Desmos will evaluate this to 22, confirming the area in square feet.

Step 3: As a sanity check, try the common error calculation as well: type 3168/12 — this gives 264, which is what you'd get if you forgot to square the conversion factor; comparing both results helps you see why squaring is necessary for area conversions.

Percentages

4 questions · Easy 0 / Medium 0 / Hard 4

Percentages — Q1 [Hard] (ID: 28ff3966)

Question ID: 28ff3966

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

A square scale drawing has a side length of 55 inches, and 1 inch on the drawing represents an actual distance of 19 miles. A larger version of the same drawing is printed as a square with the side length 65% longer than the side length of the previous drawing. On the larger drawing, which of the following is closest to the actual distance, in miles, represented by 1 inch?

Answer

A. 6.65

B. 11.52

C. 31.35

D. 35.75

Correct Answer: B

The Rule: In scale-drawing problems, the ACTUAL physical size being represented stays the same even when the drawing itself is resized — so first calculate the total actual distance represented by the original drawing, then use that same fixed actual distance to find the new scale for the resized drawing.

The Tip: Don't confuse 'the drawing got bigger' with 'the actual place got bigger' — the real-world distance being depicted stays constant; only how much space on paper represents that distance changes. Calculate the fixed real-world total first, before working with the new drawing's dimensions.

Why B is right: The original drawing's side is 55 inches, with each inch representing 19 actual miles, so the total actual distance across is 55 × 19 = 1,045 miles. This actual distance doesn't change when the drawing is resized. The new drawing's side length is 65% longer: 55 × 1.65 = 90.75 inches. Since this new, larger drawing still represents the same 1,045 actual miles, the new scale is 1,045 ÷ 90.75 ≈ 11.52 miles per inch.

Step-by-Step Solution:

Step 1: Calculate the total actual distance represented by the original drawing's side: 55 inches × 19 miles/inch = 1,045 miles.

Step 2: Recognize that this actual distance (1,045 miles) stays the SAME even when the drawing is resized — only the drawing's size changes, not the real place it represents.

Step 3: Calculate the new drawing's side length: the new side is 65% LONGER than the original 55 inches, so multiply by 1.65 (100% + 65% = 165%, or 1.65 as a decimal): 55 × 1.65 = 90.75 inches.

Step 4: Since the new 90.75-inch side still represents the same 1,045 actual miles, calculate the new scale by dividing: 1,045 ÷ 90.75 ≈ 11.5152 miles per inch.

Step 5: Rounding to match the answer choices, this is closest to 11.52, matching choice B.

The Trap: Choice C (31.35) is the most commonly picked wrong answer — it typically results from incorrectly multiplying the ORIGINAL scale (19 miles/inch) by 1.65 directly, as if the scale itself grows by 65% (19 × 1.65 ≈ 31.35), rather than recognizing that a LARGER drawing of the SAME place actually needs a SMALLER (not larger) miles-per-inch scale.

POE — Eliminate the other three:

✗ **A —** 6.65 comes from dividing 19 by something close to 1.65 backwards, or another inverted calculation — this doesn't match the correct scale-recalculation approach.

✗ **C —** 31.35 results from multiplying the original scale (19) by 1.65, treating the SCALE itself as growing — but a bigger drawing of the same place actually needs a SMALLER scale (fewer miles per inch), not a bigger one.

✗ **D —** 35.75 doesn't correspond to a clean derivation from this problem — likely an arithmetic slip or distractor value.

Solving with Desmos:

Step 1: Open Desmos and calculate the fixed actual distance first: type 55*19 — this confirms 1,045 miles is the actual distance represented by the original drawing's side.

Step 2: Calculate the new drawing's side length: type 55*1.65 — this confirms 90.75 inches for the larger drawing.

Step 3: Calculate the new scale by dividing the FIXED actual distance by the NEW side length: type 1045/90.75 — Desmos will evaluate this to approximately 11.5152, confirming the new scale is closest to 11.52 miles per inch, matching choice B.

Percentages — Q2 [Hard] (ID: 8bb94668)

Question ID: 8bb94668

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

The speeds of objects A, B, and C are a meters per second, b meters per second, and c meters per second, respectively. If the speed of object A is $5{,}900\%$ of the speed of object C and the speed of object C is 0.008% of the speed of object B, which expression represents the value of $a+b$ in terms of c?

Answer

A. $1{,}840c$

B. $5{,}908c$

C. $6{,}025c$

D. $12{,}559c$

Correct Answer: D

The Rule: When a problem describes percentages chained together across multiple variables (A is a percent of C, and C is a percent of B), convert EACH percentage into a decimal multiplier first, then use substitution to express everything in terms of a single variable.

The Tip: Convert every percentage to its decimal form immediately (5,900% becomes 59, and 0.008% becomes 0.00008) — working with percentages over 100% or under 1% in their original form is a common source of decimal-point errors; converting early keeps the rest of the algebra clean.

Why D is right: Since a is 5,900% of c, convert to decimal: a = 59c. Since c is 0.008% of b, convert to decimal: c = 0.00008b, which means b = c / 0.00008 = 12,500c. Adding: a + b = 59c + 12,500c = 12,559c.

Step-by-Step Solution:

Step 1: Convert 'a is 5,900% of c' into an equation: 5,900% as a decimal is 5900/100 = 59, so a = 59c.

Step 2: Convert 'c is 0.008% of b' into an equation: 0.008% as a decimal is 0.008/100 = 0.00008, so c = 0.00008b.

Step 3: Solve this second equation for b (since we need b in terms of c to add it to a): b = c / 0.00008.

Step 4: Calculate this division: 1/0.00008 = 12,500, so b = 12,500c.

Step 5: Now add a and b, both now expressed in terms of c: a + b = 59c + 12,500c.

Step 6: Combine like terms: 59c + 12,500c = 12,559c.

Step 7: This matches choice D.

The Trap: Choice A (1,840c) is the most commonly picked wrong answer — it typically results from a decimal-point error while converting 0.008% to a decimal, such as using 0.08 instead of the correct 0.00008, which throws off the entire calculation for b.

POE — Eliminate the other three:

✗ **A —** 1,840c likely results from a decimal-point misplacement while converting 0.008% — using too large a decimal value for this tiny percentage significantly changes the resulting value of b.

✗ **B —** 5,908c doesn't correctly combine both relationships — possibly only accounting for one of the two given percentage relationships instead of both.

✗ **C —** 6,025c also doesn't align with the correct chain of calculations — likely from a different decimal or arithmetic slip somewhere in converting one of the two percentages.

Solving with Desmos:

Step 1: Open Desmos and set up the relationships directly. First type: a=59c to represent 'a is 5,900% of c.'

Step 2: Type: c=0.00008b to represent 'c is 0.008% of b.'

Step 3: To find b in terms of c, type: b=c/0.00008 — Desmos will show this simplifies to b = 12500c (you can verify by typing 1/0.00008 separately, which evaluates to 12500).

Step 4: Finally, type a+b using the c-based expressions: 59c+12500c — Desmos will confirm this combines to 12559c, matching choice D. You can also plug in a specific test value for c (like c=1) to see the numeric results directly: with c=1, a=59 and b=12500, so a+b=12559.

Percentages — Q3 [Hard] (ID: 6feae9c3)

Question ID: 6feae9c3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

At a recreation center, visitors used the exercise room for a total of t hours in July. At the same recreation center, visitors used the exercise room for $3.49t$ hours in August. What is the percent increase in the number of hours visitors used the exercise room from July to August?

Answer

A. 2.49%

B. 3.49%

C. 249%

D. 349%

Correct Answer: C

The Rule: Percent increase is always calculated as (the amount of increase) divided by (the ORIGINAL, starting value), then converted to a percentage by multiplying by 100 — never divide by the new/final value.

The Tip: When both the original and new values are given as multiples of the same variable (like t and 3.49t here), the variable will always cancel out completely in the percent-increase calculation — don't be intimidated by having a variable instead of a specific number; just follow the formula exactly as you would with numbers.

Why C is right: The increase in hours is 3.49t − t = 2.49t. The percent increase is (increase / original) × 100 = (2.49t / t) × 100 = 2.49 × 100 = 249%.

Step-by-Step Solution:

Step 1: Identify the original value (July hours): t.

Step 2: Identify the new value (August hours): 3.49t.

Step 3: Calculate the amount of increase: new value − original value = 3.49t − t = 2.49t.

Step 4: Apply the percent increase formula: (increase / original value) × 100%.

Step 5: Substitute the values: (2.49t / t) × 100%.

Step 6: The t in the numerator and denominator cancels out completely: 2.49 × 100% = 249%.

Step 7: This matches choice C.

The Trap: Choice B (3.49%) is the most commonly picked wrong answer — it mistakes the MULTIPLIER (3.49, representing the NEW total as a multiple of the original) for the percent INCREASE itself, without first subtracting 1 (or subtracting the original amount) to isolate just the increase.

POE — Eliminate the other three:

✗ A — 2.49% correctly identifies the coefficient of the increase (2.49) but forgets to convert this decimal into a percentage by multiplying by 100 — leaving it 100 times too small.

✗ B — 3.49% confuses the total NEW amount's multiplier (3.49, meaning August is 3.49 times July) with the percent INCREASE — but a multiplier of 3.49 actually represents a 249% increase (since 3.49 = 1 + 2.49, meaning 100% of the original PLUS a 249% increase), not a 3.49% increase.

✗ D — 349% incorrectly uses the full multiplier 3.49 converted directly to a percentage (349%) without first subtracting the original 100% (representing the unchanged portion) to isolate just the increase.

Solving with Desmos:

Step 1: Since this problem involves a variable (t) that cancels out, pick a simple test value to make it concrete — let's use t = 100 (100 hours in July). Step 2: Open Desmos and calculate August's hours: type 3.49*100 — this gives 349 hours in August.

Step 3: Calculate the increase: type 349-100 — this gives 249 hours of increase.

Step 4: Calculate the percent increase using the original (July) value as the base: type (249/100)*100 — Desmos will show this evaluates to 249, confirming a 249% increase and matching choice C.

Step 5: Try this with a different test value for t (like t = 50) to confirm the percentage stays the same regardless of which specific value you pick — this confirms the variable genuinely cancels out in the percent-increase formula.

Percentages — Q4 [Hard] [Grid-In] (ID: 98818acf)

Question ID: 98818acf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

The number of audiobooks in a library increased by 131% from last year to this year. The number of audiobooks in the library last year was n, and the number of audiobooks in the library this year is bn, where b and n are constants. What is the value of b?

Correct Answer: 2.31 (Student-Produced Response — no answer choices given)

The Rule: When a quantity increases by p%, the new total is always (100% + p%) of the original — expressed as a decimal multiplier, this is (1 + p/100). Never use just p/100 alone as the multiplier, since that would only represent the AMOUNT of increase, not the new TOTAL.

The Tip: Whenever you see 'increased by X%' and need to find a multiplier that gives you the NEW TOTAL directly (not just the increase amount), always add 1 to X/100 — this combines the original 100% (which is 1, in decimal form) with the X% increase.

Why this is the answer: Since the number of audiobooks increased by 131% from last year (n) to this year, the new total is the original 100% PLUS the 131% increase, totaling 231% of the original, or 2.31 as a decimal multiplier. Since this year's count is given as bn, this means b = 2.31.

Step-by-Step Solution:

Step 1: Understand what 'increased by 131%' means: the new amount equals the original amount (100%) PLUS an additional 131% increase, for a total of 100% + 131% = 231%.

Step 2: Convert 231% into a decimal multiplier: 231/100 = 2.31.

Step 3: Since the new amount is bn (b times the original n), and we just determined the new amount is 2.31 times the original, this means b = 2.31.

Step 4: This is a grid-in question — enter 2.31 as your final answer.

The Trap: A common error on this problem is using 1.31 instead of 2.31 — this happens when a student converts 131% directly to a decimal (1.31) without recognizing that this decimal already represents the FULL new total's relationship to the original as an increase multiplier, but forgetting that 'increased BY 131%' means adding this to the original 100%, or alternatively, using just 0.31 or 1.31 by misunderstanding which part represents the total versus just the increase.

Solving with Desmos:

Step 1: Since this problem involves the variable n, pick a simple test value to make it concrete — let's use n = 100 (100 audiobooks last year). Step 2: Open Desmos and calculate the increase amount: type 100*1.31 — this gives 131, representing the SIZE of the increase (131% of 100 is 131 additional audiobooks).

Step 3: Calculate this year's total by adding the increase to the original: type 100+131 — this gives 231, the total number of audiobooks this year.

Step 4: Since this year's count is bn, and n = 100, solve for b: type 231/100 — Desmos will show this equals 2.31, confirming b = 2.31 and matching your grid-in answer.

One-variable data: distributions and measures of center and spread

4 questions · Easy 1 / Medium 1 / Hard 2

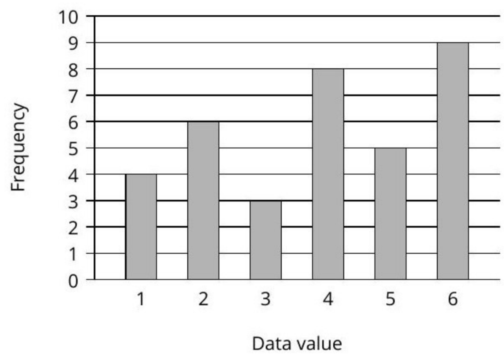
One-variable data: distributions and measures of center and spread — Q1 **[Easy]** **[Grid-In]** (ID: 559476f4)

Question ID: 559476f4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question

The bar graph shows the frequency of each data value in a data set.



What is the frequency of data value 2 in this data set?

Correct Answer: 6 (Student-Produced Response — no answer choices given)

The Rule: On a bar graph, the frequency of any specific data value is read directly from the HEIGHT of that value's bar, measured against the vertical (frequency) axis.

The Tip: Trace straight up from the data value you're asked about on the horizontal axis until you hit the top of its bar, then trace straight across (left) to read the exact height on the vertical axis — this two-step 'trace up, then trace across' method prevents misreading a neighboring bar.

Why this is the answer: Looking at the bar directly above the data value '2' on the horizontal axis, its height reaches exactly 6 on the frequency (vertical) axis.

Step-by-Step Solution:

Step 1: Locate '2' on the horizontal (data value) axis.

Step 2: Trace straight up from that point until you reach the top of the bar directly above it.

Step 3: From the top of that bar, trace straight across (left) to the vertical (frequency) axis to read its height.

Step 4: The bar for data value 2 reaches a height of 6.

Step 5: This is a grid-in question — enter 6 as your final answer.

The Trap: A common error on this type of problem is misreading an adjacent bar (like the bar for data value 1, which has height 4, or data value 3, which has height 3) instead of carefully tracing up from exactly '2' on the horizontal axis.

Solving with Desmos:

This is a straightforward graph-reading question that doesn't require Desmos to solve — the answer (6) is read directly from the bar graph provided in the question. However, if you want to double check your reading, you could recreate the bar heights as a list in Desmos (typing the data values 1 through 6 and their corresponding frequencies 4, 6, 3, 8, 5, 9 as a table) and confirm that the frequency listed for data value 2 is indeed 6.

One-variable data: distributions and measures of center and spread — Q2 [Medium] (ID: d73a67d7)

Question ID: d73a67d7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Medium

Question

Which of the following lists represents a data set with the smallest standard deviation?

Answer

A. 51, 52, 54, 56, 57

B. 52, 52, 54, 56, 56

C. 52, 53, 54, 55, 56

D. 53, 54, 54, 54, 55

Correct Answer: D

The Rule: Standard deviation measures how SPREAD OUT a data set is from its mean — a data set with values tightly clustered close together (especially with repeated identical values) will always have a SMALLER standard deviation than one with values spread far apart, even if both sets share the same mean.

The Tip: You don't need to calculate the exact standard deviation for problems like this — just visually compare how 'spread out' or 'clustered together' each list's values are. The list with values closest together (and most repeated identical values) will have the smallest standard deviation.

Why D is right: List D (53, 54, 54, 54, 55) has values clustered extremely tightly around 54, with three of the five values being EXACTLY 54 — this tight clustering around the mean gives it the smallest spread, and therefore the smallest standard deviation, compared to the other lists.

Step-by-Step Solution:

Step 1: Calculate (or estimate) the mean of each list — notice all four lists happen to have a mean of 54.

Step 2: Since all means are equal, compare how far each list's individual values deviate from 54 — the list with the SMALLEST deviations overall will have the smallest standard deviation.

Step 3: List A (51,52,54,56,57): deviations of −3,−2,0,2,3 — fairly spread out.

Step 4: List B (52,52,54,56,56): deviations of −2,−2,0,2,2 — moderately spread out.

Step 5: List C (52,53,54,55,56): deviations of −2,−1,0,1,2 — more tightly clustered.

Step 6: List D (53,54,54,54,55): deviations of −1,0,0,0,1 — extremely tightly clustered, with three values exactly AT the mean.

Step 7: List D has by far the smallest deviations from the mean, meaning it has the smallest standard deviation.

Step 8: This matches choice D.

The Trap: Choice C is the most commonly picked wrong answer — its values are evenly and closely spaced (52 through 56, each one apart), which LOOKS tightly clustered, but list D's heavy repetition of the exact mean value (54 appearing three times) actually creates even less spread than C's evenly-spaced-but-still-varying values.

POE — Eliminate the other three:

✗ A — List A has the widest range of values (51 to 57, a spread of 6) among all four lists, giving it the LARGEST standard deviation, not the smallest.

✗ B — List B has a moderate spread (52 to 56), larger than both C and D — its values aren't as tightly clustered around the mean as list D's.

✗ C — List C has evenly spaced values with no repeats, giving it a real but moderate spread — list D's heavy repetition of the exact mean (54 three times) creates even less variability than C's evenly distributed values.

Solving with Desmos:

Step 1: Open Desmos and enter each list as a set of points to calculate statistics. For list D, type: stdev({53,54,54,54,55})

Step 2: Press Enter — Desmos will calculate and display the exact standard deviation for this list.

Step 3: Repeat this same process for lists A, B, and C: stdev({51,52,54,56,57}), stdev({52,52,54,56,56}), and stdev({52,53,54,55,56})

Step 4: Compare all four calculated standard deviations directly — you'll see list D's value is clearly the smallest of the four, confirming choice D without needing to estimate visually.

One-variable data: distributions and measures of center and spread — Q3 [Hard] (ID: f6c18a66)

Question ID: f6c18a66

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

For data set A, the table summarizes the distribution of the number of deliveries received by an office each day during a period of 11 days.

Deliveries	Days
0	2
3	2
4	2
5	2
6	1
7	1
13	1

The data value 13 was recorded in error and is removed from data set A to create data set B, which consists of the remaining 10 data values. Which statement best compares the median of data set A and the median of data set B?

Answer

A. The median of data set B is less than the median of data set A.

B. The median of data set B is greater than the median of data set A.

C. The median of data set B is equal to the median of data set A.

D. There is not enough information to compare the medians of the two data sets.

Correct Answer: C

The Rule: The median of a data set is the MIDDLE value when all values are arranged in order (for an odd number of values), or the AVERAGE of the two middle values (for an even number of values). Removing a value that's far from the middle (like an outlier or error) often doesn't change the median at all.

The Tip: When a data set is described using a frequency table (value paired with how many times it occurs), first 'unroll' it into a full ordered list before trying to find the median — this makes it much easier to correctly count to the middle position(s), especially for larger data sets.

Why C is right: Data set A (11 values, unrolled from the table): 0, 0, 3, 3, 4, 4, 5, 5, 6, 7, 13. With 11 values, the median is the 6th value: counting in order, this is 4. After removing the erroneous 13, data set B (10 values): 0, 0, 3, 3, 4, 4, 5, 5, 6, 7. With 10 values (even), the median is the average of the 5th and 6th values: both are 4, so the median is (4+4)/2 = 4. Since both medians equal 4, they are equal.

Step-by-Step Solution:

Step 1: 'Unroll' the frequency table into a full ordered list for data set A: two 0's, two 3's, two 4's, two 5's, one 6, one 7, one 13 — giving: 0, 0, 3, 3, 4, 4, 5, 5, 6, 7, 13 (11 values total).

Step 2: Find the median of data set A: with 11 values (an odd number), the median is the middle (6th) value. Counting through the ordered list: 1st=0, 2nd=0, 3rd=3, 4th=3, 5th=4, 6th=4. The median of set A is 4.

Step 3: Create data set B by removing the erroneous value 13: 0, 0, 3, 3, 4, 4, 5, 5, 6, 7 (10 values total).

Step 4: Find the median of data set B: with 10 values (an even number), the median is the average of the two middle values (the 5th and 6th). Counting through: 5th=4, 6th=4. The median of set B is (4+4)/2 = 4.

Step 5: Compare the two medians: both data set A and data set B have a median of 4 — they are EQUAL.

Step 6: This matches choice C.

The Trap: Choice A is the most commonly picked wrong answer — students sometimes assume that removing the largest value (13) from a data set must automatically lower the median, without actually counting through the ordered list to check whether 13 was anywhere near the middle position in the first place (it wasn't — it was the very last, most extreme value).

POE — Eliminate the other three:

✗ A — Removing 13 (the LARGEST value, at the very end of the ordered list) doesn't affect which values sit in the MIDDLE of the data set — the median stays the same, not lower.

✗ B — Similarly, there's no reason removing the largest, most extreme value would cause the median to increase — the middle values of the data set are unaffected by removing an extreme outlier.

✗ D — Since the data set is fully known and given in the table (not an estimate or a sample), there IS enough information to precisely calculate and compare both medians — this choice incorrectly claims otherwise.

Solving with Desmos:

Step 1: Open Desmos and enter data set A as a full list based on the frequency table: median({0,0,3,3,4,4,5,5,6,7,13})

Step 2: Press Enter — Desmos will calculate and display the median, confirming it equals 4.

Step 3: On the next line, enter data set B (with 13 removed): median({0,0,3,3,4,4,5,5,6,7})

Step 4: Press Enter — Desmos will show this median is also 4, confirming both medians are equal and verifying choice C.

One-variable data: distributions and measures of center and spread — Q4 [Hard] (ID: cf2be18d)

Question ID: cf2be18d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

$$a, 26, 29, b, 31, 47, c$$

For the given data set, the data values are listed in ascending order, where a, b, and c are constants. For this data set, the mean is 36, the median is 29, and the range is 72. What is the value of c?

Answer

A. 54

B. 72

C. 81

D. 98

Correct Answer: C

The Rule: When a data set has unknown values mixed with known ones, use the GIVEN statistics (mean, median, range) as separate equations — the mean gives you a SUM equation, the median tells you which position (or unknown) sits in the middle, and the range gives you a MAX-MINUS-MIN equation. Solve this small system of equations together.

The Tip: Since the data set is already given in ascending order, use that order to immediately identify which position corresponds to the median (the exact middle value for an odd-count list) — this often directly tells you the value of one of the unknowns without any calculation.

Why C is right: With 7 values in ascending order (a, 26, 29, b, 31, 47, c), the median is the 4th value, which is b. Since the median is given as 29, this means b = 29. The mean is 36, so the sum of all 7 values is 7 × 36 = 252. Substituting known values: a + 26 + 29 + 29 + 31 + 47 + c = 252, which simplifies to a + c + 162 = 252, so a + c = 90. The range is 72, and since the list is in ascending order, the range equals the largest value minus the smallest: c − a = 72. Now solve the system: a + c = 90 and c − a = 72. Adding these two equations: 2c = 162, so c = 81.

Step-by-Step Solution:

Step 1: Since the 7 values are listed in ascending order (a, 26, 29, b, 31, 47, c), identify the median's position: for 7 values, the median is the 4th (middle) value, which is b.

Step 2: Since the median is given as 29, set b = 29.

Step 3: Use the mean to find the sum of all values: mean × count = sum, so 36 × 7 = 252.

Step 4: Substitute all known values into the sum equation: a + 26 + 29 + 29 + 31 + 47 + c = 252.

Step 5: Add up the known numbers: 26 + 29 + 29 + 31 + 47 = 162. The equation becomes: a + c + 162 = 252.

Step 6: Solve for a + c: a + c = 252 − 162 = 90.

Step 7: Use the range (largest minus smallest): since the list is ascending, the range is c − a = 72.

Step 8: Now solve this system of two equations: a + c = 90 and c − a = 72. Add both equations together: (a+c) + (c−a) = 90 + 72, which simplifies to 2c = 162.

Step 9: Divide both sides by 2: c = 81.

Step 10: This matches choice C.

The Trap: Choice A (54) is the most commonly picked wrong answer — it typically results from solving only one of the two equations (like using just the range equation with an incorrect assumed value for a) without properly setting up and solving the full system using both the mean AND range information together.

POE — Eliminate the other three:

✗ **A —** 54 doesn't satisfy both the sum equation (a+c=90) and the range equation (c−a=72) simultaneously — it likely comes from an incomplete calculation using only one piece of given information.

✗ **B —** 72 is actually the value of the RANGE itself (given in the problem), not the value of c — a student might mistakenly select the given range number instead of solving for c.

✗ **D —** 98 doesn't satisfy the system of equations either — if c = 98, then from c − a = 72, a would be 26, but then a + c = 26 + 98 = 124, which doesn't match the required sum of 90.

Solving with Desmos:

Step 1: Open Desmos and set up the two equations from the problem as a system. First type: a+c=90

Step 2: Type the second equation: c-a=72

Step 3: Desmos will treat these as two lines if you use x and y instead — replace a with x and c with y: x+y=90 and y-x=72

Step 4: Press Enter after each — Desmos graphs both lines and shows their intersection point when you click on it.

Step 5: The intersection point will show x = 9 (the value of a) and y = 81 (the value of c), confirming c = 81 and matching choice C.

Two-variable data: models and scatterplots

3 questions · Easy 1 / Medium 2 / Hard 0

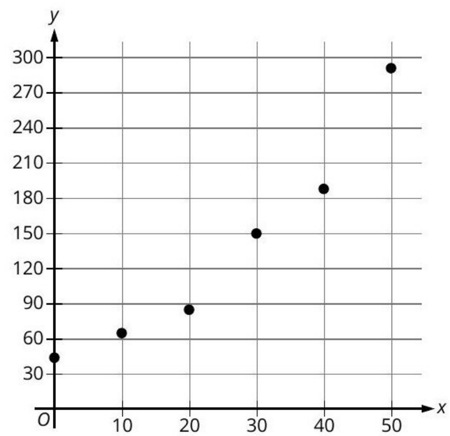
Two-variable data: models and scatterplots — Q1 **[Easy]** (ID: dbd89c59)

Question ID: dbd89c59

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Easy

Question

The scatterplot shows the relationship between the number of swans, y, in a certain population and the number of years since 1970, x, where $0 \leq x \leq 50$.



Which of the following equations is the most appropriate exponential model for the data shown in the scatterplot?

Answer

A. $y = 43(1.04)^{-x}$

B. $y = 43(1.04)^{x}$

C. $y = 288(1.04)^{-x}$

D. $y = 288(1.04)^{x}$

Correct Answer: B

The Rule: For an exponential model y = a(b)^x, the constant 'a' (the coefficient out front) represents the STARTING value (the y-intercept, where x = 0), and the base 'b' tells you whether the data is growing (b > 1) or decaying (b < 1). Match both features to the scatterplot's visual pattern.

The Tip: Always check the y-intercept FIRST on an exponential scatterplot — it should closely match the leading coefficient 'a' in the correct equation. Then check whether the data is rising or falling overall to determine if the base should be greater than or less than 1.

Why B is right: The scatterplot shows data starting at approximately y = 43 when x = 0, and clearly INCREASING (growing) as x increases up to 50. This matches a = 43 (the starting value) and a base greater than 1 with a POSITIVE exponent (1.04)^x, giving y = 43(1.04)^x.

Step-by-Step Solution:

Step 1: Read the y-intercept directly from the scatterplot: the leftmost point (at x = 0) sits at approximately y = 43.

Step 2: Observe the overall trend of the data: as x increases from 0 to 50, y clearly increases (grows) from about 43 up to nearly 290 — this is GROWTH, not decay.

Step 3: Match these observations to the exponential model form y = a(b)^x: the starting value a should be approximately 43, and since the data is growing, the exponent should be POSITIVE x (not negative x, which would represent decay).

Step 4: This matches choice B: y = 43(1.04)^x.

The Trap: Choice A is the most commonly picked wrong answer — it uses the correct starting value (43) but a NEGATIVE exponent (−x), which would actually make the function DECREASE as x increases, contradicting the clearly increasing pattern shown in the scatterplot.

POE — Eliminate the other three:

✗ **A —** The negative exponent (−x) would make this function DECREASE as x increases, but the scatterplot clearly shows the data values GROWING over time — this doesn't match the visual trend.

✗ **C —** Starting at 288 doesn't match the scatterplot's y-intercept, which is clearly close to 43 (the point at x=0), not near 288 — this starting value belongs to a very different curve.

✗ **D —** Similarly, this uses 288 as the starting value, which doesn't match the scatterplot's actual y-intercept of approximately 43 — the growth direction (positive exponent) is correct, but the starting point is wrong.

Solving with Desmos:

Step 1: Open Desmos and type the candidate equation: y=43(1.04)^x

Step 2: Press Enter — Desmos graphs this curve.

Step 3: Compare this graph to the scatterplot given in the question — check that the curve passes close to the plotted points, starting near (0, 43) and rising to approximately (50, 290).

Step 4: If the curve fits the scattered points well, you've confirmed choice B. You can also try graphing the other choices (like y=288(1.04)^x) to see how poorly they fit the actual data points, reinforcing why B is the best model.

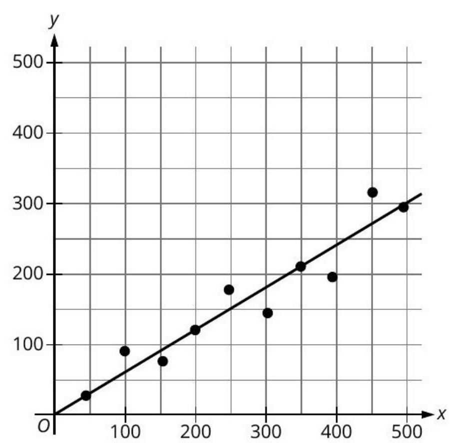
Two-variable data: models and scatterplots — Q2 [Medium] (ID: b3547a10)

Question ID: b3547a10

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Medium

Question

A group of 10 botanists recorded data on the germination rates of a certain type of seed for one growing season. The scatterplot shows the relationship between the number of seeds planted, x, and the number of seeds that germinated, y, for each of the botanists. A line of best fit is also shown.



Which of the following is the best interpretation of the slope of the line of best fit in this context?

Answer

A. The number of seeds planted is predicted to increase by 60 seeds every 100 days.

B. The number of seeds planted is predicted to increase by 300 seeds every 100 days.

C. The number of seeds that germinate is predicted to increase by 60 seeds for every additional 100 seeds that are planted.

D. The number of seeds that germinate is predicted to increase by 300 seeds for every additional 100 seeds that are planted.

Correct Answer: C

The Rule: The slope of a line of best fit represents the RATE OF CHANGE between the two variables graphed — specifically, how much the DEPENDENT variable (on the y-axis) changes for every one-unit increase in the

INDEPENDENT variable (on the x-axis). Always identify which variable is on which axis before interpreting the slope in context.

The Tip: Read the axis labels carefully before interpreting slope — the correct interpretation must have the y-axis variable as the one that 'increases' and the x-axis variable as the one changing 'per' some amount. Answer choices that swap these two roles are testing whether you're reading the graph correctly.

Why C is right: On this scatterplot, x represents the number of seeds PLANTED, and y represents the number of seeds that GERMINATED. The slope of the line of best fit, estimated from the graph (rising from about (0,30) to (500,310), giving a slope of about 280/500 = 0.56, closest to 60/100 = 0.6), means that for every 100 additional seeds planted, approximately 60 more seeds are predicted to germinate.

Step-by-Step Solution:

Step 1: Identify which variable is on the x-axis (independent/input variable) and which is on the y-axis (dependent/output variable): x = number of seeds planted, y = number of seeds that germinated.

Step 2: Estimate two points on the line of best fit from the graph: approximately (0, 30) and (500, 310).

Step 3: Calculate the slope using these two points: slope = (310 − 30) / (500 − 0) = 280/500 = 0.56.

Step 4: Compare this slope to the answer choices, which are given as 'X seeds per 100 seeds': 0.56 is closest to 60/100 = 0.60.

Step 5: Since the slope relates y (germinated) to x (planted), the correct interpretation must describe germinated seeds increasing per planted seeds — not the reverse.

Step 6: This matches choice C: 'The number of seeds that germinate is predicted to increase by 60 seeds for every additional 100 seeds that are planted.'

The Trap: Choice A is the most commonly picked wrong answer — it has the correct approximate slope value (60 per 100) but describes the WRONG variable as increasing (seeds planted, the independent/x-axis variable, rather than seeds germinated, the dependent/y-axis variable) — this reverses the cause-and-effect direction the slope actually represents.

POE — Eliminate the other three:

✗ A — This describes 'seeds planted' as the variable that increases — but seeds planted is the INDEPENDENT variable (x-axis), which the researchers choose directly; it's seeds GERMINATED (y-axis) that responds to and increases with planting, not the reverse.

✗ B — This has the same variable-reversal issue as A, plus uses the wrong slope magnitude (300 instead of approximately 60) — doesn't match the graph's actual slope.

✗ D — This correctly identifies germinated seeds as the increasing variable but uses the wrong slope magnitude (300 per 100, which would be a much steeper line than what's shown) — the actual slope is closer to 60 per 100, not 300.

Solving with Desmos:

Step 1: Open Desmos and estimate several points from the scatterplot to check the line of best fit's slope. Type a few points you can read from the graph, like: (100,90), (200,120), (350,215), (450,315)

Step 2: Desmos will plot these points. To find a best-fit line, click the '+' button and look for a regression feature, or manually type: y1~mx+b and Desmos will calculate the best-fit slope (m) and intercept (b) automatically using linear regression.

Step 3: Check the calculated slope value — it should be close to 0.56 to 0.6, confirming that for every 100 seeds planted (x), about 60 seeds are predicted to germinate (y), matching choice C's interpretation.

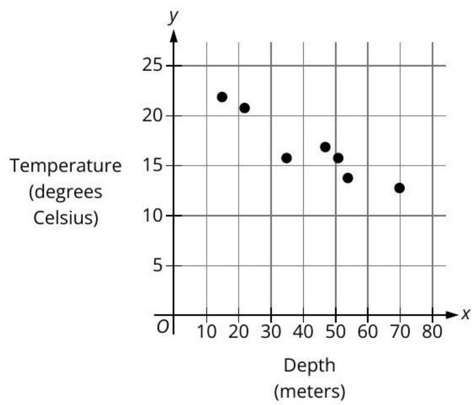
Two-variable data: models and scatterplots — Q3 [Medium] (ID: 70a2776f)

Question ID: 70a2776f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Medium

Question

As part of a study of lake conditions, the water temperature was recorded at different depths below the surface of a certain lake. The data shown in the scatterplot give the recorded temperature, in degrees Celsius, for 7 depths, in meters, below the surface of the lake.



Which of the following is closest to the slope of a line of best fit for the data shown?

Answer

A. -0.17

B. -2.17

C. -6.02

D. -8.02

Correct Answer: A

The Rule: To estimate the slope of a line of best fit from a scatterplot, choose two points that appear to represent the OVERALL trend of the data (not just two random points), calculate the slope between them using (change in y) / (change in x), and match this to the closest answer choice.

The Tip: When estimating slope from a scatterplot with a downward trend, pick one point from the far LEFT side of the data and one from the far RIGHT side — using points that are far apart minimizes the impact of any single point's imprecise reading, giving a more reliable slope estimate.

Why A is right: Using an early point around (15, 22) and a later point around (70, 13), the slope is estimated as (13 − 22) / (70 − 15) = −9/55 ≈ −0.16, which is closest to −0.17 among the answer choices.

Step-by-Step Solution:

Step 1: Choose two points that represent the overall trend of the scattered data — one from early depths and one from later depths. Reading the graph, approximate points are (15, 22) and (70, 13).

Step 2: Calculate the change in y (temperature): 13 − 22 = −9.

Step 3: Calculate the change in x (depth): 70 − 15 = 55.

Step 4: Calculate the slope: −9 / 55 ≈ −0.1636.

Step 5: Compare this estimated slope to the answer choices: −0.1636 is closest to −0.17.

Step 6: This matches choice A.

The Trap: Choice B (−2.17) is the most commonly picked wrong answer — it's off by roughly a factor of 10-13 from the correct slope, which can happen if a student misreads the axis scale (for example, confusing gridline spacing) while estimating the change in x or y.

POE — Eliminate the other three:

✗ **B** — −2.17 is far too steep for this data — this magnitude of slope would mean temperature drops over 150 degrees across the full depth range shown, which doesn't match the actual data (which only spans about 22 to 13 degrees).

✗ **C** — −6.02 is even more drastically too steep — this doesn't match the gentle downward trend visible in the scattered points.

✗ **D** — −8.02 is also far too steep for the gentle decline shown in the data — likely from a significant misreading of the graph's scale or points.

Solving with Desmos:

Step 1: Open Desmos and enter several points estimated from the scatterplot: (15,22), (20,21), (35,16), (45,17), (50,16), (52,14), (70,13)

Step 2: Desmos will plot these points. To find the actual best-fit slope, type: y1~mx+b — Desmos will use linear regression to calculate the best-fit values for m (slope) and b (intercept) automatically.

Step 3: Check the calculated value of m — it should be close to −0.17, confirming your estimate and matching choice A.

Probability and conditional probability

2 questions · Easy 0 / Medium 0 / Hard 2

Probability and conditional probability — Q1 [Hard] [Grid-In] (ID: 89ff6a0a)

Question ID: 89ff6a0a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	Hard

Question

For 100 buttons, the table summarizes the distribution by group and diameter.

	Diameter (millimeters)		
Group	Less than 20	20 to 30	Greater than 30
Group 1	12	8	3
Group 2	0	17	18
Group 3	10	32	0

One of these buttons will be selected at random. What is the probability of selecting a button with a diameter that is less than or equal to 30 millimeters, given that it is **not** in group 2? (Express your answer as a decimal or fraction, not as a percent.)

Correct Answer: 62/65 (Student-Produced Response — no answer choices given)

The Rule: For conditional probability problems presented in a table ('probability of X given Y'), first identify and total ONLY the rows/columns that satisfy the GIVEN condition (the 'given that' part), then find what fraction of that reduced total also satisfies the main event being asked about.

The Tip: Physically cross out or ignore any table cells that DON'T meet the 'given that' condition before calculating anything — this prevents accidentally including irrelevant data in either your numerator or denominator.

Why this is the answer: 'Given that it is NOT in group 2' means we only consider Group 1 and Group 3 (ignoring Group 2 entirely). Group 1 total: 12+8+3 = 23. Group 3 total: 10+32+0 = 42. Combined total (not group 2): 23+42 = 65. Within this reduced group, 'diameter ≤ 30mm' means the 'less than 20' and '20 to 30' columns: Group 1: 12+8=20; Group 3: 10+32=42. Combined: 20+42 = 62. The probability is 62/65.

Step-by-Step Solution:

Step 1: Identify the condition ('given that it is NOT in group 2'): this means we only look at Group 1 and Group 3 rows, completely ignoring Group 2.

Step 2: Calculate the total number of buttons in Group 1: 12 + 8 + 3 = 23.

Step 3: Calculate the total number of buttons in Group 3: 10 + 32 + 0 = 42.

Step 4: Calculate the combined total for 'not in group 2': 23 + 42 = 65. This is our new denominator.

Step 5: Now find how many of these 65 buttons have a diameter ≤ 30mm — this means the 'Less than 20' AND '20 to 30' columns (both are ≤ 30mm), but NOT the 'Greater than 30' column.

Step 6: From Group 1: 'Less than 20' = 12, '20 to 30' = 8. Sum: 12 + 8 = 20.

Step 7: From Group 3: 'Less than 20' = 10, '20 to 30' = 32. Sum: 10 + 32 = 42.

Step 8: Combine these: 20 + 42 = 62. This is our numerator.

Step 9: The probability is 62/65 (this fraction cannot be simplified further, since 62 = 2×31 and 65 = 5×13 share no common factors).

Step 10: This is a grid-in question — enter 62/65 (or its decimal equivalent, approximately 0.954) as your final answer.

The Trap: A common error on this problem is using the TOTAL of all 100 buttons as the denominator (instead of just the 65 buttons that are 'not in group 2'), which incorrectly calculates a regular probability instead of the required CONDITIONAL probability.

Solving with Desmos:

Step 1: Open Desmos as a calculator to organize this conditional probability calculation. Type: 12+8+3 to confirm Group 1's total is 23.

Step 2: Type: 10+32+0 to confirm Group 3's total is 42.

Step 3: Type: 23+42 to confirm the 'not group 2' total is 65 (your denominator).

Step 4: Type: 12+8+10+32 to confirm the count with diameter ≤ 30mm within 'not group 2' is 62 (your numerator).

Step 5: Type: 62/65 — Desmos will show this as approximately 0.9538, confirming your fraction and allowing you to enter either 62/65 or the decimal into the grid-in box.

Probability and conditional probability — Q2 [Hard] (ID: f496d2c0)

Question ID: f496d2c0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	Hard

Question

The table shows the distribution of rooms in a certain facility by seating capacity.

Seating capacity	Proportion
Less than 18 seats	26%
18–40 seats	21%
41–65 seats	29%
Greater than 65 seats	24%

If a room in this facility is selected at random, which of the following is closest to the probability of selecting a room that has a seating capacity greater than 65 seats, given that the room has a seating capacity of at least 18 seats?

Answer

A. 0.24

B. 0.32

C. 0.50

D. 0.92

Correct Answer: B

The Rule: For conditional probability problems with percentages, first add up ONLY the percentages that satisfy the GIVEN condition to find the new total, then divide the specific percentage you're looking for by this new (reduced) total — not by 100%.

The Tip: Rewrite the 'given that' condition as its own smaller, self-contained group first — add up just the relevant percentages to get a new 'whole' for that group, then calculate the requested probability as a fraction of THAT smaller whole, not the original 100%.

Why B is right: 'Given that the room has a seating capacity of at least 18 seats' means combining the '18–40 seats,' '41–65 seats,' and 'greater than 65 seats' categories: 21% + 29% + 24% = 74%. Within this 74% group, the probability of 'greater than 65 seats' is 24% ÷ 74% ≈ 0.3243, which rounds to approximately 0.32.

Step-by-Step Solution:

Step 1: Identify the given condition: 'seating capacity of at least 18 seats' — this includes the '18–40 seats,' '41–65 seats,' and 'greater than 65 seats' categories (everything except 'less than 18 seats').

Step 2: Add up the percentages for these three categories to find the new total: 21% + 29% + 24% = 74%.

Step 3: Identify what we're looking for within this reduced group: rooms with 'greater than 65 seats,' which is 24%.

Step 4: Calculate the conditional probability by dividing the specific percentage by the new total: 24% / 74% ≈ 0.3243.

Step 5: Rounding to match the answer choices, this is closest to 0.32.

Step 6: This matches choice B.

The Trap: Choice A (0.24) is the most commonly picked wrong answer — it uses the original 24% directly as the answer, forgetting to divide by the reduced total (74%) that accounts for the 'given that' condition — this would be the correct answer only if the question asked for the UNCONDITIONAL probability across all rooms, not the conditional probability.

POE — Eliminate the other three:

✗ A — 0.24 is simply the original percentage for 'greater than 65 seats' without adjusting for the 'given that' condition — this ignores the conditional nature of the question entirely.

✗ C — 0.50 doesn't correspond to a correct calculation from this table — likely a distractor or a significant miscalculation.

✗ D — 0.92 might come from dividing the wrong two percentages together, or from a completely different (incorrect) combination of the given values.

Solving with Desmos:

Step 1: Open Desmos and calculate the new total for the 'given' condition: type 21+29+24 — this confirms 74% is the combined total for rooms with at least 18 seats.

Step 2: Calculate the conditional probability: type 24/74 — Desmos will evaluate this to approximately 0.3243.

Step 3: Compare this decimal to the answer choices — 0.3243 rounds to 0.32, confirming choice B.

Inference from sample statistics and margin of error

3 questions · Easy 0 / Medium 2 / Hard 1

Inference from sample statistics and margin of error — Q1 **[Medium]** (ID: 3e1bd4e2)

Question ID: 3e1bd4e2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	Medium

Question

Researchers studied a sample of a species of fish from a certain location. The researchers found that the mean length of fish in the sample was 16.59 millimeters. Based on the sample mean and margin of error, the researchers estimated that the mean length of all fish of this species from this location was between 15.86 and 17.32 millimeters. What is the margin of error, in millimeters?

Answer

A. 0.73

B. 2.19

C. 15.86

D. 16.59

Correct Answer: A

The Rule: A confidence interval (the range given for an estimate) is always constructed as (sample estimate) ± (margin of error). This means the margin of error is exactly HALF the total width of the interval, or equivalently, the distance from the sample mean to either endpoint of the interval.

The Tip: To find the margin of error from a given interval, you don't need the full formula — simply subtract the sample mean from the UPPER endpoint (or subtract the LOWER endpoint from the sample mean); both calculations give you the same margin of error value.

Why A is right: The sample mean is 16.59mm, and the interval's upper bound is 17.32mm. The margin of error is the difference: $17.32 - 16.59 = 0.73$. (You can double check: the lower bound is $16.59 - 0.73 = 15.86$, which matches the given lower bound exactly.)

Step-by-Step Solution:

Step 1: Identify the sample mean (the central estimate): 16.59 millimeters.

Step 2: Identify one endpoint of the confidence interval — let's use the upper bound: 17.32 millimeters.

Step 3: Calculate the margin of error by finding the difference between the upper bound and the sample mean: $17.32 - 16.59 = 0.73$.

Step 4: Double-check using the lower bound: sample mean − margin of error should equal the lower bound: $16.59 - 0.73 = 15.86$, which matches the given lower bound exactly, confirming the calculation.

Step 5: This matches choice A.

The Trap: Choice C (15.86) is the most commonly picked wrong answer — it directly uses the LOWER BOUND of the interval as if it were the margin of error itself, rather than correctly calculating the DIFFERENCE between the sample mean and that bound.

POE — Eliminate the other three:

✗ B — 2.19 doesn't correspond to a direct calculation from the given numbers — possibly from adding instead of subtracting, or another arithmetic slip.

✗ C — 15.86 is simply the LOWER BOUND of the confidence interval itself, not the margin of error — the margin of error is the DISTANCE between the mean and this bound, not the bound's value directly.

✗ D — 16.59 is simply the SAMPLE MEAN itself, not the margin of error — selecting this confuses the central estimate with the amount of uncertainty around it.

Solving with Desmos:

Step 1: Open Desmos and calculate the margin of error directly: type 17.32-16.59

Step 2: Press Enter — Desmos will evaluate this to 0.73, confirming the margin of error and matching choice A.

Step 3: As a double check, verify using the lower bound instead: type 16.59-15.86 — this should ALSO give 0.73, confirming your answer is consistent from both directions of the interval.

Inference from sample statistics and margin of error — Q2 [Medium] (ID: 2bab1399)

Question ID: 2bab1399

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	Medium

Question

A researcher is designing a study to investigate the average number of hours participants in a training course spend on a particular activity per day. The researcher will report an estimated average number of hours participants in the training course spend on this activity per day with an associated margin of error. The researcher is considering using a random sample of either 115 or 230 participants from the training course. Which of the following would be the most likely effect of using the larger random sample compared to the smaller random sample?

Answer

A. The reported margin of error would be lower.

B. The reported margin of error would be higher.

C. The reported average number of hours would be lower.

D. The reported average number of hours would be higher.

Correct Answer: A

The Rule: In statistical sampling, INCREASING the sample size always tends to DECREASE the margin of error (making the estimate more precise), while decreasing the sample size increases the margin of error. This is because larger samples give a more reliable, representative picture of the full population.

The Tip: Remember this general relationship as an inverse one: sample size UP means margin of error DOWN, and sample size DOWN means margin of error UP. This relationship holds true across virtually all standard statistical sampling methods, so you can answer these conceptual questions without any calculation.

Why A is right: Since 230 participants is a LARGER sample than 115 participants (twice as large), using this larger sample would be expected to produce a MORE precise (narrower) estimate — meaning a LOWER margin of error compared to the smaller sample.

Step-by-Step Solution:

Step 1: Compare the two sample sizes being considered: 115 participants versus 230 participants.

Step 2: Recognize that 230 is the LARGER of the two sample sizes (twice as large as 115).

Step 3: Apply the general statistical principle: larger sample sizes lead to more precise estimates, which means a SMALLER (lower) margin of error.

Step 4: Since the question asks about the effect of using the LARGER sample (230) compared to the smaller one (115), the expected effect is a LOWER margin of error.

Step 5: This matches choice A.

The Trap: Choice B is the most commonly picked wrong answer — it reverses the correct relationship, assuming that MORE data somehow leads to a HIGHER margin of error (more 'noise' or uncertainty), when in fact more data generally REDUCES uncertainty and produces a tighter, more reliable estimate.

POE — Eliminate the other three:

✗ **B** — This reverses the true relationship between sample size and margin of error — larger samples produce LOWER margins of error (more precision), not higher ones.

✗ **C** — The sample size affects the PRECISION of the estimate (the margin of error), not necessarily the reported average value itself — a larger sample doesn't systematically make the average number of hours lower.

✗ **D** — Similarly, sample size doesn't systematically make the reported average HIGHER either — it affects how precise (narrow) the margin of error is around whatever the true average happens to be.

Solving with Desmos:

This is a conceptual statistics question about the relationship between sample size and margin of error, which doesn't require a specific calculation in Desmos. However, if you want to explore this relationship numerically, margin of error in many standard formulas is proportional to $1/\sqrt{\text{sample size}}$ — you could type sqrt(115) and sqrt(230) into Desmos to compare: you'll see sqrt(230) is larger than sqrt(115), and since margin of error is roughly proportional to 1 divided by this square root, a LARGER sample size (230) corresponds to a SMALLER margin of error, confirming choice A conceptually.

Inference from sample statistics and margin of error — Q3 [Hard] (ID: da978ee6)

Question ID: da978ee6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	Hard

Question

A researcher surveyed a random sample of $1,020$ people from a certain city to estimate the percentage of people who support a recently announced proposal. From the survey, the researcher estimated that 86% of people from the city support this proposal, with an associated margin of error of 2.13%. The researcher repeated the survey with a random sample of people from the city that was double the original sample size. Assuming the margins of error were calculated in the same way, which of the following is true about the margin of error associated with the estimate from the larger sample?

Answer

A. The margin of error is between 2.13% and 3.13%.

B. The margin of error is between 4.13% and 5.13%.

C. The margin of error is greater than 5.13%.

D. The margin of error is less than 2.13%.

Correct Answer: D

The Rule: For simple random sampling, the margin of error is inversely proportional to the SQUARE ROOT of the sample size — meaning if you increase the sample size by some factor, the margin of error decreases by the square root of that same factor, not by the factor itself.

The Tip: When a sample size DOUBLES, don't assume the margin of error simply halves — because of the square-root relationship, doubling the sample size only shrinks the margin of error by a factor of $\sqrt{2}$ (about 1.41), not by a full factor of 2. Recognizing this square-root relationship (rather than a direct proportional one) is the key insight for these problems.

Why D is right: Doubling the sample size from 1,020 to 2,040 people means the margin of error decreases by a factor of $\sqrt{2} \approx 1.414$. So the new margin of error is approximately $2.13\% \div 1.414 \approx 1.506\%$, which is LESS than the original 2.13%.

Step-by-Step Solution:

Step 1: Identify the relationship between sample size and margin of error: margin of error is inversely proportional to the square root of the sample size.

Step 2: Identify the change in sample size: the new sample is DOUBLE the original ($2 \times 1,020 = 2,040$ people).

Step 3: Calculate the factor by which the margin of error changes: since sample size increased by a factor of 2, margin of error decreases by a factor of $\sqrt{2} \approx 1.414$.

Step 4: Calculate the new (approximate) margin of error: $2.13\% \div 1.414 \approx 1.506\%$.

Step 5: Compare this new margin of error (approximately 1.51%) to the original (2.13%) and to the answer choices — 1.51% is LESS than 2.13%, matching the description in choice D.

Step 6: This matches choice D.

The Trap: Choice A is the most commonly picked wrong answer — it assumes the margin of error will still be reasonably close to the original value (between 2.13% and 3.13%, even higher than before), missing that a LARGER sample size should DECREASE the margin of error, not increase or maintain it.

POE — Eliminate the other three:

✗ A — This suggests the margin of error would be HIGHER than or similar to the original (2.13% to 3.13%) — but doubling the sample size should DECREASE the margin of error, not increase it.

✗ B — This suggests an even larger margin of error (4.13% to 5.13%) — this strongly contradicts the fundamental principle that larger samples produce smaller, more precise margins of error.

✗ C — This suggests the margin of error would be extremely high (greater than 5.13%) — this is the opposite of what should happen with a larger, more representative sample size.

Solving with Desmos:

Step 1: Open Desmos and calculate the square-root scaling factor: type sqrt(2) — this confirms the factor is approximately 1.414 (since the sample size doubled).

Step 2: Calculate the new margin of error by dividing the original by this factor: type 2.13/sqrt(2)

Step 3: Press Enter — Desmos will evaluate this to approximately 1.506, confirming the new margin of error is about 1.51%, which is LESS than the original 2.13%, matching choice D.

Step 4: As a broader check, try a few other sample size ratios (like tripling the sample, using sqrt(3) instead of sqrt(2)) to build intuition for how margin of error shrinks more slowly than the sample size grows — this square-root relationship is a recurring theme in AP Statistics and college-level statistics as well.